



FIRST SEMESTER 2026-2027

Course Handout (Part II)

Date: 31st July, 2025

In addition to Part-I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No : CHEM F312
Course Title : Physical Chemistry IV
Instructor-in-charge : SHAMIK CHAKRABORTY
Instructor(s) : SHAMIK CHAKRABORTY

1. Course Description:

Weak forces; surface chemistry: interphase region, thermodynamics, surface films on liquids, adsorption of gases on solids, colloids, micelles, and reverse micellar structures; transport processes: kinetics, thermal conductivity, viscosity, diffusion, sedimentation; electrical conductivity in metals and in solutions; reaction kinetics, measurement of rates; integrated rate laws; rate laws and equilibrium constants for elementary reactions; reaction mechanisms; temperature dependence of rate constants; rate constants and equilibrium constants; rate law in non-ideal systems; uni, bi and tri molecular reactions, chain reactions, free-radical polymerizations; fast reactions; reactions in solutions; heterogeneous and enzyme catalysis; introduction to statistical mechanics; theories of reaction rates; molecular reaction dynamics.

2. Scope and Objective of the course:

This is the last one of the four Physical Chemistry courses named for M.Sc. (Hons.) Chemistry Programme. The course mainly covers chemical changes in terms of chemical kinetics, surface and interfacial phenomena, and the associated theories. The objective is to understand the chemical changes and also to be able to connect these changes to structure and equilibrium properties learnt in the previous courses.

3. Text Book:

T1: Peter Atkins and Julio de Paula, *Atkins' Physical Chemistry*, 10th Ed.(South Asia Edition), Oxford University Press, Oxford, 2014.

Reference Book :

R1. Levine Ira N., *Physical Chemistry*, 6th ed., Tata McGraw-Hill, New Delhi, 2011.

4. Course Plan:

Lecture No.	Topic	Learning Objectives	Learning Outcome	Ref. to Text Book/Ref. book
1-4	Statistical Thermodynamics	The Boltzmann distribution, Partition Functions, Molecular Energies, The Canonical ensemble.	Gaining insight into the nature of 'temperature', partition function, mean values of the basic modes of motion, explaining interaction between molecules through 'ensemble' concept	T1: 15A – 15D





5-7	“	The internal Energy and the Entropy, Derived Functions	To gain knowledge how ‘partition function bridges between ‘spectroscopy’ and ‘thermodynamics’ as well as between spectroscopy, structural data and equilibrium constant	T1: 15E – 15F
8-10	Rates of chemical reactions; Integrated rate laws; Finding of rate law.	Definition of rate, derivation of concentration time relationship for simple reactions, determination of rate law, half-life of reactions, reactions approaching equilibrium, Expt. Procedures to obtain rate laws.	Mathematical expression of reaction rate in terms of conc. of reactants and products, verification of proposed rate law, determination of order and rate constant of a reaction, relation between forward and backward rate constants with equilibrium constant.	T1: 20A – 20C
11	Temperature effects on rates	Concept of activation energy	Gaining knowledge about the minimum energy needed for a reaction to occur in an encounter between reactants	T1: 20D
12-14	Elementary reaction Mechanisms	Elementary reactions, composite reactions, steady-state approximation, rate determining step; pre-equilibria; Kinetic and Thermodynamic control of reactions	Gaining knowledge about how to construct rate law for multi-step reactions.	T1: 20E R1: 16.5 - 16.6, 16.9-16.10
15-16	Examples of Reaction Mechanisms	Unimolecular reactions, Polymerization kinetics, stepwise polymerization, chain polymerization	Using the concept of steady-state approximations in deriving rate laws of reactions that follow complex reaction mechanisms.	T1: 20F R1: 16.11 - 16.12
17-18	Photochemistry	Photochemical process, primary quantum yield, mechanism of decay of excited singlet states, Quenching, Resonance energy transfer	How to include the effect of electromagnetic radiations in the rate laws of chemical and biological processes.	T1: 20G
19-20	Enzymes	Features of Enzymes, The Michaelis-Menten Mechanism, The catalytic efficiency of enzymes, Mechanism of enzyme inhibition	The role of enzymes as bio-catalyst and how to inhibit the catalytic activities of enzymes.	T1: 20H.1 – 20H.4





21-22	Collision theory	Reactive encounters, RRK model	Collision between two molecules in gas phase leads to reaction when their relative kinetic energy exceeds a threshold value and certain steric requirements.	T1: 21A.1 – 21A.2
23-24	Diffusion controlled reactions	Reaction in solution, Material- balance equations	Reactions in solution are either diffusion controlled or activation controlled.	T1: 21B.1 – 21B.2
25-27	Transition State Theory	The Eyring equation, Thermodynamic Aspects, The Kinetic isotope effect	Gaining insight about the link between structure of reactants and rate constant for their reaction.	T1: 21C.1 – 21C.3
28-29	The dynamics of Molecular collision	Reaction trajectory; Molecular reaction Dynamics	Investigation of the rates of reactions in gas phase by exploring the trajectories of molecules on potential energy surfaces.	T1: 21D.1 – 21D.4 R1: 22.2-22.3
30-31	Electron Transfer in homogeneous catalysis	The electron transfer rate law, the rate constant	The rate constant of electron transfer in donor-acceptor complex depends on distance between donor and acceptor, the standard reaction Gibbs energy and the energy needed to reach a particular arrangement of atoms.	T1: 21E.1 – 21E.2
32-34	Molecular Interactions	Electric dipole moments, Polarizabilities, Polarization, Interaction between partial charges, Interaction between dipoles, Interaction between dipole-induced dipole, Interaction between induced dipoles, Hydrogen bonding, Hydrophobic interaction, Total attractive and repulsive interactions	Understanding of the relation between electronic structure and electric properties of molecules and how these electric properties lead to the formation of condensed phases, molecular assemblies, e.g. how proteins and nucleic acids fold into their three-dimensional structures.	T1: 16A – 16B R1: 13.14, 21.10
35-36	Liquids	Molecular interactions in liquids, Liquid-vapour interface, surface films, Thermodynamics of surface layers, Condensation	Gaining insight into the process of condensation of a gas into liquids, thermodynamic properties of liquids and formation of liquid surface.	T1: 16C R1: 7.6 - 7.8
37-38	Self-Assembly	Classification, Preparation, Structure & stability of colloids, Micelle formation,	Understanding of how colloids, micelles and biological membranes form spontaneously	T1: 17C – 17D R1: 7.9





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		Reverse micellar structures, bilayers, Determination of size & shape	by self-assembly of molecules or macromolecules and are held together by molecular interactions.	
39-40	Transport processes	The phenomenological equations, The transport parameters, Liquid viscosity, mobility of ions, diffusion	A molecule carries properties through space for about the distance of its mean free path, viscosity of a liquid decreases with increasing temperature, ions reach a thermal velocity when the electrical force on them is balanced by the drag due to the viscosity of the solvent, knowledge about spreading of one material through another when reactions takes place in solution and spreading of substances into the environment.	T1: 19A – 19C R1: 15.1 - 15.7

5. Evaluation Scheme:

Components	Duration	Marks (% weightage)	Date & Time	Venue	Remarks
Mid-Semester Test	90 mins	60 (30%)	06/10/2026 11.00 A.M. - 12.30 P.M.	To be announced by AUGSD	Closed Book/Open Book
Continuous Evaluation*	25 minutes each	40 (20%)	Continuous (best 2 out of 3 will be considered. Each test will be of 20 marks)	Allotted Classroom	Closed Book/Open Book
Continuous Evaluation**	10 mins each	20 (10%)			
Comprehensive Examination	180 mins	80 (40%)	05/12/2026 & 2.00 P.M. - 05.00 P.M.	To be announced by AUGSD	Closed Book

* In continuous evaluation, THREE tutorial tests (20 marks each) will be conducted in entire semester. Best TWO will be considered for final evaluation. All quizzes will be pre-announced.

**In continuous evaluation, linked to attendance, FOUR quizzes, each of five marks, will be conducted (TWO before MID-SEM examination and TWO after MID-SEM examination). The quiz marks (10%) will be considered for final grading only if the student maintains a minimum of 50% attendance. All quizzes will be pre-announced.

@ **Tutorials:** The tutorial hour will be used for a quick review of the highlights of the material covered in the lectures, clarification of doubts and problem-solving.

6. Chamber Consultation Hour: TBA





7. Minimum criterion for awarding valid grades: Where a student continued to remain registered in a course but gave the instructor inadequate opportunity to evaluate him/her by absenting himself/herself from quizzes/ tests/ examinations/other components of evaluation, or by appearing in the same for the sake of appearance without applying himself/herself to the task in hand or by submitting a blank script (answer book) this event will be reported as NC (Not Cleared). **Moreover, a student should obtain 30% of the average of the top 10 performers in the class, or 40% of the median marks of the class, whichever is lower, to clear the course.** If any student obtains marks below the prescribed standard mentioned above, he/she may be awarded an NC.

8. Chamber Consultation Hour: TBA

9. Makeup Policy: See Part I for details. However, it may be noted that there will be no make-up for tutorials since the **‘best three’ out of four evaluations** will only be considered.

10. Notices: Notices, if any, concerning the course will be conveyed either through e-mail or the Nalanda portal.

**Instructor-in-Charge
(CHEM F312)**

